

The V5 Finite-Memory Kernel as a Substrate-Level Resolution of the Trans-Planckian Problem

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Abstract

The standard semiclassical derivation of Hawking radiation traces observed-at-infinity modes back to arbitrarily-blueshifted near-horizon frequencies. The derivation thus implicitly requires standard QFT to apply at and beyond the Planck scale, where quantum-gravity effects should structurally modify field theory. This is the *trans-Planckian problem*, recognized since Jacobson [1] (1991) and analyzed extensively in subsequent literature [2–6]. Multiple proposals address it phenomenologically: modified dispersion relations [1, 2], lattice cutoffs [3], naturalness arguments [4], the Trans-Planckian Censorship Conjecture (TCC) [5], and others. We show that the Event Density (ED) framework’s V5 finite-memory cross-chain kernel, with characteristic memory time $\tau_{V5} = \ell_P/c$ at the gravitational scale (the natural substrate-built timescale via T19 + dimensional analysis), produces a substrate-level resolution that does not require any of the standard phenomenological additions. The substrate cannot mediate cross-chain correlations at frequencies $\omega \gg c/\ell_P$ because V5’s coherence breaks down beyond the cutoff. Combined with the Diffusion Coarse-Graining Theorem’s hydrodynamic-window closure at $L_{\text{flow}} \sim \ell_P$, the substrate naturally cuts off arbitrarily-blueshifted modes near the horizon. The standard trans-Planckian divergence is a continuum artifact that vanishes at the substrate level. The V5-modulated Hawking spectrum is $N_{ED}(\omega) = N_H(\omega)/(1 + (\omega\tau_{V5})^2)$, finite at all frequencies and reducing to standard Hawking at observable scales. Falsifiable predictions in current-generation analog Hawking experiments and primordial-BH late-stage evaporation distinguish the framework from modified-dispersion and lattice-cutoff approaches.

Keywords: trans-Planckian problem, Hawking radiation, Event Density, substrate ontology, V5 finite-memory kernel, UV regulation, modified dispersion, Diffusion Coarse-Graining Theorem.

1 Introduction

In the standard semiclassical derivation of Hawking radiation [7, 8], a Schwarzschild black hole of mass M emits thermal radiation at temperature $T_H = \kappa/(2\pi)$, where κ is the surface gravity of the horizon. The derivation runs through QFT in curved spacetime: vacuum modes propagated from past null infinity, through the BH-collapse spacetime, to future null infinity, where they appear as a thermal mixture via the Bogoliubov transformation between past and future mode bases.

The derivation has empirical content (analog Hawking experiments [9–10] confirm the spectral form) but a structural concern. Modes observed at moderate frequencies $\omega \sim T_H$ at substrate-asymptotic infinity were arbitrarily-blueshifted near the horizon at proper frequencies

$$\omega_{\text{proper}}(r) = \omega \cdot (1 - 2M/r)^{-1/2}$$

For $r \rightarrow r_h^+$ (horizon-approaching), $\omega_{\text{proper}} \rightarrow \infty$. The standard derivation thus implicitly requires standard QFT to apply at proper frequencies arbitrarily far above the Planck scale

$\omega_P = c/\ell_P \approx 1.85 \times 10^{43}$ Hz, where quantum-gravity effects should structurally modify field theory.

This is the *trans-Planckian problem* of Hawking radiation [1]. It has been addressed in the literature through several proposals:

- **Modified dispersion relations** [1, 2]: postulate UV-modified dispersion $\omega^2 = c^2 k^2 (1 + \mathcal{O}(\ell_P k))$ at the Planck scale.
- **Lattice cutoffs** [3]: discretize spacetime at the Planck scale, eliminating modes with $k \gg 1/\ell_P$.
- **Naturalness / robustness arguments** [4]: argue that physics-domain robustness should protect Hawking radiation from trans-Planckian modifications.
- **Trans-Planckian Censorship Conjecture (TCC)** [5]: conjectured cosmological censorship of trans-Planckian modes by sufficient inflation.
- **'t Hooft S-matrix arguments** [6]: structural arguments from BH information theory.

Each proposal is structurally meaningful but each is *phenomenological* in character: each posits the cutoff or modification rather than deriving one from a substrate-level ontology. The trans-Planckian problem has therefore remained a structural concern of standard semiclassical Hawking, without a derivation-level resolution.

This paper presents the Event Density (ED) framework's substrate-level resolution. The framework is built on a substrate ontology of discrete micro-events, finite participation bandwidth, irreversible commitment events, and finite-width substrate kernels [11, 12]. The framework's V5 finite-memory cross-chain kernel — a primitive substrate object with characteristic memory time τ_{V5} — naturally regulates the high-frequency behavior of substrate-mediated correlations. At the gravitational scale, $\tau_{V5} = \ell_P/c$ (the Planck time, identified via T19 + substrate dimensional analysis [13]). This is the natural substrate-built timescale at which V5 coherence breaks down.

The framework's claim: the substrate cannot mediate cross-chain correlations at frequencies $\omega \gg c/\ell_P$. Arbitrarily-blueshifted near-horizon modes do not exist at the substrate level. The trans-Planckian divergence of standard semiclassical Hawking is a *continuum artifact* of the QFT-in-curved-spacetime calculation; the substrate-level account regulates it naturally without postulating modified dispersion, lattice cutoffs, or any other phenomenological addition.

Combined with the Diffusion Coarse-Graining Theorem's (DCGT) hydrodynamic-window closure at $L_{\text{flow}} \sim \ell_P$, the framework provides a substrate-level UV regulation that:

1. Reproduces standard Hawking exactly at observable scales.
2. Cuts off the high-frequency tail at the Planck frequency.
3. Resolves the trans-Planckian problem without new physics or postulated regulators.
4. Produces falsifiable first-subleading-order corrections distinguishing the framework from standard semiclassical Hawking and from the major phenomenological resolutions.

The structure of the paper: §2 states the standard trans-Planckian problem precisely. §3 presents the V5 kernel at the substrate level. §4 derives the V5-modulated Hawking spectrum. §5 articulates DCGT's role in the resolution. §6 states the substrate-level resolution explicitly. §7 contrasts with the major standard-physics proposals. §8 identifies falsifiable predictions. §9 discusses the implications.

2 The Standard Trans-Planckian Problem

2.1 The blueshift argument

In Schwarzschild geometry, an outgoing mode observed at substrate-asymptotic infinity with frequency ω traces back to a mode near the horizon at proper frequency

$$\omega_{\text{proper}}(r) = \omega \cdot (1 - 2M/r)^{-1/2}.$$

For an observer at radius $r = 2M(1 + \epsilon)$ with small ϵ , the proper frequency is approximately $\omega/\sqrt{2\epsilon}$. As $\epsilon \rightarrow 0$, $\omega_{\text{proper}} \rightarrow \infty$. Modes traced back to the horizon at any positive distance away are blueshifted to arbitrarily high proper frequencies.

For a stellar-mass black hole observed mode at $\omega \sim T_H$ (peak of the Hawking spectrum), the proper frequency exceeds the Planck scale at distances

$$\epsilon \lesssim (T_H/\omega_P)^2 \cdot 2 \sim 10^{-86}$$

— a substantial fraction of the horizon’s neighborhood, in the relevant blueshift sense.

2.2 The structural concern

Standard QFT in curved spacetime is a continuum-level theory whose validity at and beyond the Planck scale is structurally questionable. Quantum-gravity effects should modify field theory at the Planck scale, where:

- Spacetime fluctuations become order-unity.
- Quantum-gravity corrections to dispersion relations should appear.
- The continuum description of spacetime should break down.

The trans-Planckian problem articulates this concern: the standard semiclassical Hawking calculation requires QFT to be valid at arbitrarily-high proper frequencies near the horizon, including frequencies far above where QFT’s structural status is secure.

2.3 The standard responses

Multiple proposals have been advanced:

- (a) **Modified dispersion relations** [1, 2]. Postulate that the standard relativistic dispersion $\omega^2 = c^2 k^2$ is modified at the Planck scale to $\omega^2 = c^2 k^2 + f(k\ell_P)$ for some function f that effectively cuts off mode propagation at $k \sim 1/\ell_P$. This requires postulating the specific form of f .
- (b) **Lattice cutoffs** [3]. Discretize spacetime at the Planck scale. This requires postulating the discrete spacetime structure and its specific properties.
- (c) **Naturalness / robustness arguments** [4]. Argue that the empirical robustness of Hawking radiation should protect it from trans-Planckian sensitivity. This is an argument-from-empirical-robustness rather than a derivation.
- (d) **Trans-Planckian Censorship Conjecture (TCC)** [5]. Conjecture that cosmological inflation must terminate before trans-Planckian modes have stretched to observable scales. This is a conjecture about cosmological dynamics; it does not directly resolve the BH trans-Planckian problem.
- (e) **’t Hooft S-matrix arguments** [6]. Structural arguments from BH information theory and the holographic principle, suggesting that the BH S-matrix should be unitary and structurally regulated.

Each proposal contributes structural insight; each remains phenomenological in character. The trans-Planckian problem awaits a derivation-level resolution.

3 The V5 Finite-Memory Kernel

3.1 Substrate definition

The V5 cross-chain memory kernel is a primitive substrate object in the Event Density framework [11, 12]. Its primitive structural form is determined by:

- **Forward-cone support.** V5 mediates cross-chain correlations causally forward in substrate-time (inherited from T18 [14]).
- **Finite memory.** V5 has a characteristic correlation time τ_{V5} — a substrate-determined timescale that varies by physical context.
- **Substrate locality.** V5 mediates correlations only along substrate-locality-permitted pathways.

The minimal substrate-consistent kernel form is:

$$V_5(t) = \mathcal{V}_0 \cdot \theta(t) \cdot e^{-t/\tau_{V5}} \cdot \psi(t/\tau_{V5})$$

where $\theta(t)$ is the Heaviside step function (forward-cone-only), $e^{-t/\tau_{V5}}$ is the leading-order decay envelope, $\psi(t/\tau_{V5})$ is a substrate-determined dimensionless shape function approaching unity at small argument, and $\mathcal{V}_0$ is the substrate-coupling normalization. The exact functional form of ψ is INHERITED from substrate-microscopic details and is not fixed at the framework's structural-foundations level.

3.2 Frequency-domain structure

The Fourier transform of the simplest substrate-consistent V5 form ($\psi = 1$):

$$\tilde{V}_5(\omega) = \frac{\mathcal{V}_0 \tau_{V5}}{1 - i\omega \tau_{V5}}$$

The magnitude squared, governing V5's coherent transfer of cross-chain correlations:

$$|\tilde{V}_5(\omega)|^2 = \frac{(\mathcal{V}_0 \tau_{V5})^2}{1 + (\omega \tau_{V5})^2}$$

For $\omega \ll \omega_c \equiv 1/\tau_{V5}$, $|\tilde{V}_5(\omega)|^2$ is approximately constant — V5 is effectively local-in-time and behaves as a delta-function memory.

For $\omega \gtrsim \omega_c$, $|\tilde{V}_5(\omega)|^2 \rightarrow (1/\omega^2)$ — V5 coherence breaks down. The substrate cannot mediate cross-chain correlations at frequencies above the cutoff scale.

3.3 The cutoff scale at the gravitational scale

The cutoff τ_{V5} is a substrate-determined parameter. At the gravitational scale, the only substrate-built timescale available from substrate primitives is the Planck time:

$$\tau_{V5} = \ell_P / c \approx 5.4 \times 10^{-44} \text{ s}$$

This identification is FORCED via T19 (Newton-recovery ℓ_P) [13] plus substrate dimensional analysis. The Planck time is the natural substrate timescale at which V5 coherence breaks down at gravitational scales.

The corresponding cutoff frequency:

$$\omega_c = c/\ell_P \approx 1.85 \times 10^{43} \text{ Hz}$$

— the Planck frequency. The V5 kernel does not coherently mediate cross-chain correlations at proper frequencies approaching the Planck scale.

3.4 Why τ_{V5} is not phenomenological

The V5 cutoff scale is *not* a phenomenological cutoff inserted ad hoc to regulate the trans-Planckian problem. The V5 kernel is a substrate primitive in its own right — it appears across multiple framework sectors:

- Soft-matter Maxwell viscoelasticity, where τ_{V5} identifies with molecular relaxation times [15].
- Hawking radiation high-frequency spectral cutoff (this paper).
- Black-hole information transfer bandwidth modulation [16, 17].
- Substrate-Unruh radiation cutoff [18].

The kernel exists *before* and *independently of* the trans-Planckian problem. The cutoff scale at the gravitational context is FORCED by substrate dimensional analysis. The trans-Planckian regulation is a *consequence* of V5’s primitive structure, not a feature added to address the problem.

4 The V5-Modulated Hawking Spectrum

4.1 The leading-order substrate Hawking calculation

The framework’s leading-order substrate-level Hawking calculation [19, 20] reproduces the standard semiclassical result via the substrate-Unruh argument applied at a saturated decoupling surface (substrate-level analog of a BH horizon). V5 cross-chain correlations across the surface, satisfying imaginary-time periodicity at $\beta = 2\pi/\kappa_{ED}$ (forced by substrate-vacuum self-consistency at the surface), produce a Planck spectrum at temperature $T = \kappa_{ED}/(2\pi)$. DCGT identification at leading order gives $T_H = \kappa/(2\pi)$ — the standard Hawking temperature.

The leading-order spectrum:

$$N_H(\omega) = \frac{1}{e^{\omega/T_H} - 1}$$

This is the standard result, exactly reproduced.

4.2 First-subleading-order V5 modulation

At first subleading order, V5’s finite-memory structure modulates the spectrum. The substrate-level near-horizon correlation function involves V5 cross-chain mediation at frequency ω . The leading-order calculation treats V5 as effectively local-in-time (valid for $\omega \ll \omega_c$). The first-subleading correction includes the explicit V5 frequency-domain modulation:

$$N_{ED}(\omega) = N_H(\omega) \cdot \frac{|\tilde{V}_5(\omega)|^2}{|\tilde{V}_5(0)|^2} = \frac{N_H(\omega)}{1 + (\omega\tau_{V5})^2}$$

This is the V5-modulated Hawking spectrum. For modes with $\omega \ll \omega_c = c/\ell_P$:

$$N_{ED}(\omega) \approx N_H(\omega) \cdot [1 - (\omega\tau_{V5})^2 + \mathcal{O}((\omega\tau_{V5})^4)]$$

The modulation is approximately unity, and the spectrum reduces to the standard Planck distribution.

For modes with $\omega \gg \omega_c$:

$$N_{ED}(\omega) \approx N_H(\omega) \cdot (\omega_c/\omega)^2$$

The modulation suppresses the high-frequency tail. The substrate-level spectrum is finite and well-defined at all frequencies.

4.3 The peak of the spectrum

The Hawking spectrum's peak is at $\omega \sim T_H \cdot k_B / \hbar$. For a Schwarzschild stellar-mass BH:

$$T_H \approx 6 \times 10^{-8} \text{ K}, \quad \omega_{\text{peak}} \sim 10^4 \text{ Hz}$$

The corresponding $\omega_{\text{peak}} \tau_{V5} \sim 10^4 \cdot 10^{-43} \sim 10^{-39}$ — invisible at the spectrum peak.

For a primordial BH in its final stages of evaporation ($M \rightarrow M_P$):

$$T_H \rightarrow T_P \sim 10^{32} \text{ K}, \quad \omega_{\text{peak}} \sim \omega_c$$

The V5 cutoff becomes order-unity at the peak. The spectrum is dramatically modified from standard Hawking at extreme scales.

4.4 Integrated emission rate

The framework's integrated Page emission rate is also modulated by V5 [21]:

$$\left. \frac{dM}{dt} \right|_{ED} = -\frac{\alpha_{\text{Page}}}{M^2} \cdot \left[1 - K \left(\frac{\ell_P}{M} \right)^2 + \mathcal{O} \left(\frac{\ell_P}{M} \right)^4 \right]$$

where $K = -c_{V5}^{\text{int}} - c_g^{\text{int}} \log g$ is an order-unity coefficient inherited from V5 and motif-alphabet substrate details. The correction is invisible at observable BH scales but becomes order-unity at $M \sim M_P$.

5 DCGT and the Closing Hydrodynamic Window

The V5 cutoff is one structural component of the framework's substrate-level resolution. The Diffusion Coarse-Graining Theorem (DCGT) [15] supplies the second.

5.1 The DCGT scale-separation requirement

DCGT establishes the substrate-to-continuum bridge through hydrodynamic-window scale separation:

$$\ell_P \ll R_{\text{cg}} \ll L_{\text{flow}}$$

The substrate-level lower bound is ℓ_P (the substrate's irreducible length scale). The upper bound L_{flow} is the characteristic length scale of the continuum theory under derivation. For Hawking radiation, L_{flow} is set by the BH's size: $L_{\text{flow}} = r_s = 2M$.

The scale separation requires $M \gg \ell_P$ substantially. For ordinary BHs ($M \gg M_P$), the window is enormous (ratios of 10^{38} to 10^{60} are typical). For Planck-mass-approaching BHs ($M \sim M_P$), the window closes.

5.2 The window closure and its meaning

When M approaches ℓ_P , the upper and lower bounds of the hydrodynamic window coincide. There is no scale R_{cg} satisfying $\ell_P \ll R_{\text{cg}} \ll 2M$. The DCGT substrate-to-continuum bridge fails.

What does this mean structurally? Hawking radiation as a continuum-level phenomenon depends on the DCGT bridge. The bridge's leading-order consequences (scalar diffusion, directional viscosity, V1→R1 substrate-cutoff, V5→Maxwell viscoelasticity, T17 minimal coupling [15]) all require the hydrodynamic-window separation. When the window closes, the substrate-level dynamics no longer produce continuum-level emission as a coherent process.

For the trans-Planckian problem specifically: blueshifted near-horizon modes at proper frequencies $\omega_{\text{proper}} \gtrsim 1/\ell_P$ have proper wavelengths $\lambda_{\text{proper}} \lesssim \ell_P$. These modes lie *below* the DCGT hydrodynamic window’s lower bound. They are not continuum-level features at all.

5.3 The combined substrate-level cutoff

The two structural components combine to provide the substrate-level cutoff:

1. **V5 finite-memory cutoff (perturbative).** The V5 kernel’s coherent response saturates at $\omega \sim 1/\tau_{V5} = c/\ell_P$. Substrate cross-chain correlations at frequencies above the cutoff are suppressed by $1/\omega^2$. This is the leading-order substrate-cutoff structure.
2. **DCGT hydrodynamic-window closure (non-perturbative).** When proper wavelengths approach ℓ_P , the substrate-to-continuum bridge fails entirely. The continuum description does not apply. This is the structural-foundational substrate-cutoff statement.

The two components are complementary. V5 provides a smooth perturbative cutoff at frequencies below the DCGT failure scale. DCGT provides the structural-foundational statement that the continuum description does not extend into the trans-Planckian regime.

6 The Substrate-Level Resolution

6.1 Statement of the resolution

The substrate-level trans-Planckian resolution. Substrate cross-chain correlations at frequencies $\omega \gg c/\ell_P$ are not coherently mediated. The substrate’s V5 finite-memory kernel does not support coherent correlations beyond $\tau_{V5}^{-1} = c/\ell_P$. Combined with DCGT’s hydrodynamic-window closure when proper wavelengths approach ℓ_P , the substrate provides a natural UV cutoff at the Planck frequency. Modes at proper frequencies above the Planck scale do not exist as continuum-level objects. The standard trans-Planckian divergence of semiclassical Hawking is a *continuum artifact* of extending QFT beyond its substrate-level support; the substrate-level account regulates the divergence naturally without postulating modified dispersion, lattice cutoffs, or any phenomenological addition.

6.2 Why this is not a phenomenological cutoff

The framework’s resolution is structurally distinct from postulating an ad-hoc cutoff:

- The V5 kernel is a substrate primitive that exists *independently* of the trans-Planckian problem. The same kernel produces Maxwell viscoelastic memory in soft matter and entanglement-bandwidth modulation in BH information transfer [16].
- The cutoff scale $\tau_{V5} = \ell_P/c$ at the gravitational scale is FORCED by T19 + substrate dimensional analysis. It is the unique substrate-built timescale at the gravitational context.
- DCGT’s hydrodynamic-window closure is a structural feature of the substrate-to-continuum bridge, not a regulator inserted to address the trans-Planckian problem.

The trans-Planckian regulation is therefore a *consequence* of substrate ontology, not a feature added to address a structural concern. This is the methodological distinction between phenomenological regularization and substrate-level resolution.

6.3 What this delivers

The framework’s resolution:

1. Reproduces standard Hawking exactly at observable scales.
2. Produces a finite, well-defined Hawking spectrum at all frequencies.

3. Cuts off the high-frequency tail at the Planck frequency via V5.
4. Establishes the structural-foundational cutoff via DCGT closure when wavelengths approach ℓ_P .
5. Predicts substrate-cutoff corrections at first subleading order $(\ell_P/M)^2$.
6. Combined with H-8's higher-order resummation [21], produces a stable Planck-mass remnant at the late-time evaporation endpoint.

The resolution is therefore not just a regularization — it is a substrate-level account of where, why, and how the standard semiclassical framework's structural concerns are resolved.

7 Comparison with Standard Proposals

7.1 Modified dispersion relations (Unruh, Jacobson, BMPS)

Modified-dispersion approaches [1, 2] postulate UV-modified dispersion relations such as:

$$\omega^2 = c^2 k^2 + f(k\ell_P)$$

with $f(k\ell_P)$ a function that effectively cuts off mode propagation at $k \sim 1/\ell_P$. Specific forms include $f(k\ell_P) = -(c\ell_P k)^4/c^2$ (Unruh-class), and various lattice-derived forms.

Comparison with ED:

- *Modified dispersion*: postulates the dispersion modification as a phenomenological feature.
- *ED*: derives the high-frequency cutoff from V5's substrate primitive structure. The dispersion relation itself is unchanged at the leading order; the modification appears as a multiplicative modulation of the spectrum, not as a modification of the underlying field-theoretic mode structure.

Empirical distinguishability: Modified-dispersion approaches typically predict specific UV modification forms that depend on the chosen $f(k\ell_P)$. ED's V5 modulation $1/(1 + (\omega\tau_{V5})^2)$ is form-FORCED by V5's substrate primitive. Precision analog Hawking experiments could distinguish between different cutoff functional forms.

7.2 Lattice cutoffs

Lattice-cutoff approaches [3] discretize spacetime at the Planck scale, eliminating modes with wavelengths shorter than the lattice spacing.

Comparison with ED:

- *Lattice cutoff*: discretizes spacetime structure at the Planck scale by postulate.
- *ED*: substrate ontology is *already* discrete (P01: discrete micro-events). The substrate ontology underlying ED is not a regularization of continuum spacetime but the framework's primitive ontology. The continuum description emerges as a coarse-grained leading-order consequence (DCGT [15]).

Structural distinction: ED's discreteness is ontologically primary; lattice cutoff approaches treat discreteness as a regulator added to a continuum-level theory. In ED, the continuum is the emergent description; in lattice approaches, the continuum is the fundamental description being regulated.

7.3 Naturalness and robustness arguments

Naturalness arguments [4] argue that Hawking radiation should be *robust* against trans-Planckian sensitivities — that the empirically validated content (the thermal spectrum at $T_H = \kappa/(2\pi)$) is protected from quantum-gravity modifications by general arguments about effective field theory.

Comparison with ED:

- *Robustness arguments*: argue from empirical-protection that trans-Planckian content shouldn't matter, without supplying a mechanism.
- *ED*: provides the explicit mechanism (V5 substrate-cutoff + DCGT closure) by which trans-Planckian modes are absent at the substrate level.

Methodological distinction: Robustness arguments are arguments-from-empirical-protection; ED supplies the underlying mechanism that makes the empirical robustness structurally inevitable.

7.4 Trans-Planckian Censorship Conjecture (TCC)

TCC [5] conjectures that cosmological inflation must terminate before trans-Planckian modes have stretched to observable scales. This addresses cosmological trans-Planckian sensitivity but not directly the BH trans-Planckian problem.

Comparison with ED:

- *TCC*: cosmological constraint on inflation duration.
- *ED*: substrate-level UV cutoff applicable in any context (BH, cosmological horizon, accelerated observer [18]).

Scope distinction: TCC and ED's V5 cutoff address different aspects of the trans-Planckian problem. TCC concerns cosmological inflation; ED's V5 cutoff applies to all substrate-level modes. The two could in principle coexist; ED's V5 mechanism does not require TCC.

7.5 Firewalls and information-paradox-class proposals

Firewall [22] and related proposals (ER=EPR [23], soft hair [24], island formula [25]) address BH information at the horizon scale but do not directly resolve the trans-Planckian problem. They address what happens *at* or *near* the horizon, not the structural question of whether QFT applies at trans-Planckian frequencies.

Comparison with ED: ED addresses a structurally different question. Firewalls / ER=EPR / soft hair / island formula address the BH information question (handled separately in [16]). The trans-Planckian problem is addressed by V5 + DCGT, independently of the BH information mechanism.

7.6 Comparison summary

Approach	Mechanism	Status	Empirical distinguishability
Modified dispersion	Postulated UV dispersion modification	Phenomenological	Specific cutoff form depends on chosen f
Lattice cutoffs	Postulated spacetime discreteness	Phenomenological regulator	Discrete-spectrum signatures at small scales
Naturalness arguments	Empirical-protection argument	Argumentative	Cannot directly distinguish
TCC	Cosmological constraint on inflation	Conjectural	Cosmological inflation observables
ED V5 + DCGT	Substrate primitive + structural closure	Substrate-derived	Form-FORCED $1/(1 + (\omega\tau)^2)$ cutoff

8 Falsifiable Predictions

The framework’s substrate-level resolution distinguishes itself from standard-physics approaches through specific falsifiable predictions.

8.1 Analog Hawking cutoff form

In analog Hawking experiments [9, 10], the Hawking-analog spectrum should exhibit a high-frequency cutoff at the analog system’s microscopic correlation timescale. The framework predicts:

- The cutoff form is $1/(1 + (\omega\tau_{\text{analog}})^2)$.
- The analog timescale τ_{analog} identifies with the analog system’s characteristic correlation time:
 - BEC: $\tau_{\text{analog}} \sim \xi/c_s$ (healing length over speed of sound).
 - Acoustic: $\tau_{\text{analog}} \sim \lambda_{\text{phonon}}/c_s$.
 - Photonic: $\tau_{\text{analog}} \sim \text{optical-cycle period}$.

Existing analog Hawking experiments confirm spectral form at moderate frequencies; precision tests at the cutoff scale are technically feasible with current-generation experiments. Modified-dispersion approaches predict specific cutoff functional forms depending on the chosen $f(k\ell_P)$; ED’s prediction is the specific V5-derived $1/(1 + (\omega\tau)^2)$ form.

8.2 Primordial BH late-stage spectra

For primordial black holes in their final stages of evaporation, the framework predicts:

- Spectral cutoff at $\omega \sim \omega_c = c/\ell_P$ (Planck frequency).
- Modified evaporation profile slowing in the final ~ 0.1 s.
- Possible Planck-mass remnant after evaporation halts (per H-8 [21]).

Standard semiclassical Hawking predicts complete evaporation in finite time with no upper-frequency cutoff. PBH detection with sufficient temporal and spectral resolution would test the framework’s substrate-cutoff prediction.

8.3 High-energy photon dispersion in vacuum

If the V5 substrate-cutoff at $\omega_c = c/\ell_P$ extends to vacuum-substrate propagation (not only to BH horizon contexts), high-energy astrophysical photons traveling astronomical distances could exhibit dispersion at order $(\omega/\omega_c)^2$:

$$\Delta v/c \sim (\omega/\omega_c)^2$$

Public LIGO/Virgo + Fermi-LAT data on high-energy gamma-ray bursts and gravitational-wave-electromagnetic counterparts permit precision time-of-flight measurements. This connects ED’s V5 substrate-cutoff to the active observational program on Lorentz-invariance-violation searches at high energies [26].

8.4 Cross-platform consistency

The V5 kernel does multiple substrate-level jobs across the framework’s other domains:

- Soft-matter Maxwell viscoelasticity at molecular relaxation timescales.
- Substrate-Unruh radiation cutoff at accelerated-observer Rindler-like horizons.
- Black-hole information transfer bandwidth modulation.

Cross-platform measurement of V5-mediated phenomena across these distinct domains provides a structural consistency test. If the V5 kernel’s primitive structural form $1/(1+(\omega\tau)^2)$ is correctly identified in all four domains, the substrate-ontology evidence accumulates from cross-domain reach. If the kernel’s form differs across domains, the substrate-level identification needs revision.

8.5 Distinctive predictions versus standard proposals

Prediction	Modified dispersion	Lattice cut-off	Naturalness	TCC	ED V5 + DCGT
Specific cutoff form	Depends on $f(k\ell_P)$	Lattice-spacing-dep.	No specific prediction	Cosmo-only	$1/(1+(\omega\tau)^2)$ FORM-FORCED
Cross-platform consistency	Not predicted	Not predicted	Not predicted	Not predicted	τ_{V5} identification across domains
Late-stage PBH evaporation	Mod.-disp.-dep.	Lattice-cutoff-dep.	Robust prediction	No prediction	Planck-mass remnant + spectrum cutoff
Connection to soft matter	None	None	None	None	Maxwell relaxation time identification

9 Discussion

9.1 The structural-resolution claim

The trans-Planckian problem of standard semiclassical Hawking has been recognized for over thirty years [1] and addressed through multiple phenomenological approaches. Each approach contributes structural insight; none has produced a derivation-level resolution from a substrate-level ontology.

The Event Density framework’s V5 + DCGT account is structurally distinct. The V5 kernel is a substrate primitive existing for reasons independent of the trans-Planckian problem (cross-domain duties in soft matter, BH information, substrate-Unruh). The DCGT bridge is the framework’s substrate-to-continuum machinery. Their combination produces a substrate-level UV cutoff at the Planck scale as a *consequence* of substrate ontology, not as a feature added to address the trans-Planckian concern.

This is what we call a *substrate-level resolution*. It differs from phenomenological regularization in that the regulator is not introduced for the problem; it emerges from the substrate-level account that produces the framework’s other structural commitments.

9.2 What this does not claim

The substrate-level resolution does *not* claim:

- That standard semiclassical Hawking is wrong. It correctly describes the empirically validated content (thermal spectrum at T_H at observable scales).
- That ED is the unique substrate-level framework capable of producing such a resolution. Other substrate ontologies might in principle produce analogous resolutions.
- That all features of standard Hawking are reproduced exactly. First-subleading-order corrections at $(\ell_P/M)^2$ distinguish ED from strict semiclassical Hawking, particularly at extreme scales.

What it *does* claim is structural: the substrate ontology produces the trans-Planckian regulator as a derived consequence rather than as a phenomenological addition.

9.3 Implications for substrate-ontology evidence

The cross-platform reach of the V5 kernel — operating in soft-matter rheology, Hawking radiation, BH information, and substrate-Unruh contexts [16] — provides structural evidence for the substrate ontology. A phenomenological framework would have separate cutoff parameters for each context; ED has one substrate primitive (V5) doing all jobs, with the cutoff scale τ_{V5} inheriting from the physical context.

The trans-Planckian resolution illustrates this pattern. The same V5 kernel that produces Maxwell viscoelastic memory in soft matter at nanosecond timescales also produces the trans-Planckian cutoff at the Planck scale. The cross-domain reach is structural evidence that V5 is a primitive substrate object, not a domain-specific phenomenological regulator.

9.4 Open questions

The substrate-level resolution leaves several questions for future work:

1. **Closed-form derivation of τ_{V5} at the gravitational scale.** The framework identifies $\tau_{V5} = \ell_P/c$ via dimensional analysis; closed-form derivation from substrate-microscopic V1 + V5 details would tighten the identification.
2. **The V5 kernel shape function ψ .** Different shape functions produce different higher-order corrections; precision tests could distinguish.
3. **Analog Hawking precision tests.** Current analog experiments confirm spectral form at moderate frequencies; precision measurements at the cutoff scale would test ED’s specific functional form against modified-dispersion alternatives.
4. **Connection to the Standard Model UV completion question.** The V5 substrate cutoff regulates trans-Planckian Hawking modes; whether it also regulates the Standard Model’s UV-completion concerns at the Planck scale is open.

These open items do not undermine the substrate-level resolution but identify directions for continued development.

10 Conclusions

The trans-Planckian problem of standard semiclassical Hawking radiation arises because the standard derivation traces observed-at-infinity modes back to arbitrarily-blueshifted near-horizon proper frequencies. The standard derivation thus implicitly requires standard QFT to apply at and beyond the Planck scale, where structural concerns about quantum-gravity modification arise.

The Event Density framework provides a substrate-level resolution. The V5 finite-memory cross-chain kernel — a primitive substrate object with characteristic memory time $\tau_{V5} = \ell_P/c$ at the gravitational scale (FORCED via T19 + substrate dimensional analysis) — does not coherently mediate cross-chain correlations at frequencies $\omega \gg c/\ell_P$. Combined with the Diffusion Coarse-Graining Theorem’s hydrodynamic-window closure when proper wavelengths approach ℓ_P , the substrate provides a natural UV cutoff at the Planck frequency. The substrate cannot instantiate arbitrarily-blueshifted modes near the horizon; the trans-Planckian divergence of standard semiclassical Hawking is a continuum artifact that vanishes at the substrate level.

The V5-modulated Hawking spectrum is:

$$N_{ED}(\omega) = \frac{1}{e^{\omega/T_H} - 1} \cdot \frac{1}{1 + (\omega\tau_{V5})^2}$$

finite at all frequencies, reducing to standard Hawking at observable scales, and cut off at the Planck frequency.

The framework’s resolution is structurally distinct from standard-physics approaches:

- Modified-dispersion approaches postulate UV modification of the dispersion relation.
- Lattice-cutoff approaches postulate spacetime discreteness at the Planck scale.
- Naturalness arguments invoke empirical robustness without supplying a mechanism.
- TCC addresses cosmological inflation, not the BH trans-Planckian problem.

ED’s V5 + DCGT mechanism produces the cutoff as a *derived consequence* of substrate ontology rather than as a phenomenological addition. The V5 kernel exists for reasons independent of the trans-Planckian problem (cross-domain duties in soft matter, BH information, and substrate-Unruh). The trans-Planckian regulation is a cross-platform application of the same substrate primitive.

Falsifiable predictions distinguishing the framework include the V5-derived cutoff form $1/(1 + (\omega\tau)^2)$ in analog Hawking experiments, the Planck-frequency cutoff in primordial-BH late-stage evaporation, possible high-energy photon dispersion at order $(\omega/\omega_c)^2$, and cross-platform identification of τ_{V5} across the framework’s V5-mediated phenomena.

The substrate-level resolution illustrates a general methodological pattern: substrate primitives composed with domain-specific substrate context produce continuum-level observables in apparently unrelated domains, providing structural evidence for the substrate ontology through cross-domain reach.

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31. The full Event Density corpus is available at <https://github.com/allen-proxmire/event-density>.